

Tumor tracking in real-time 2D MRI with out-the-box vision models

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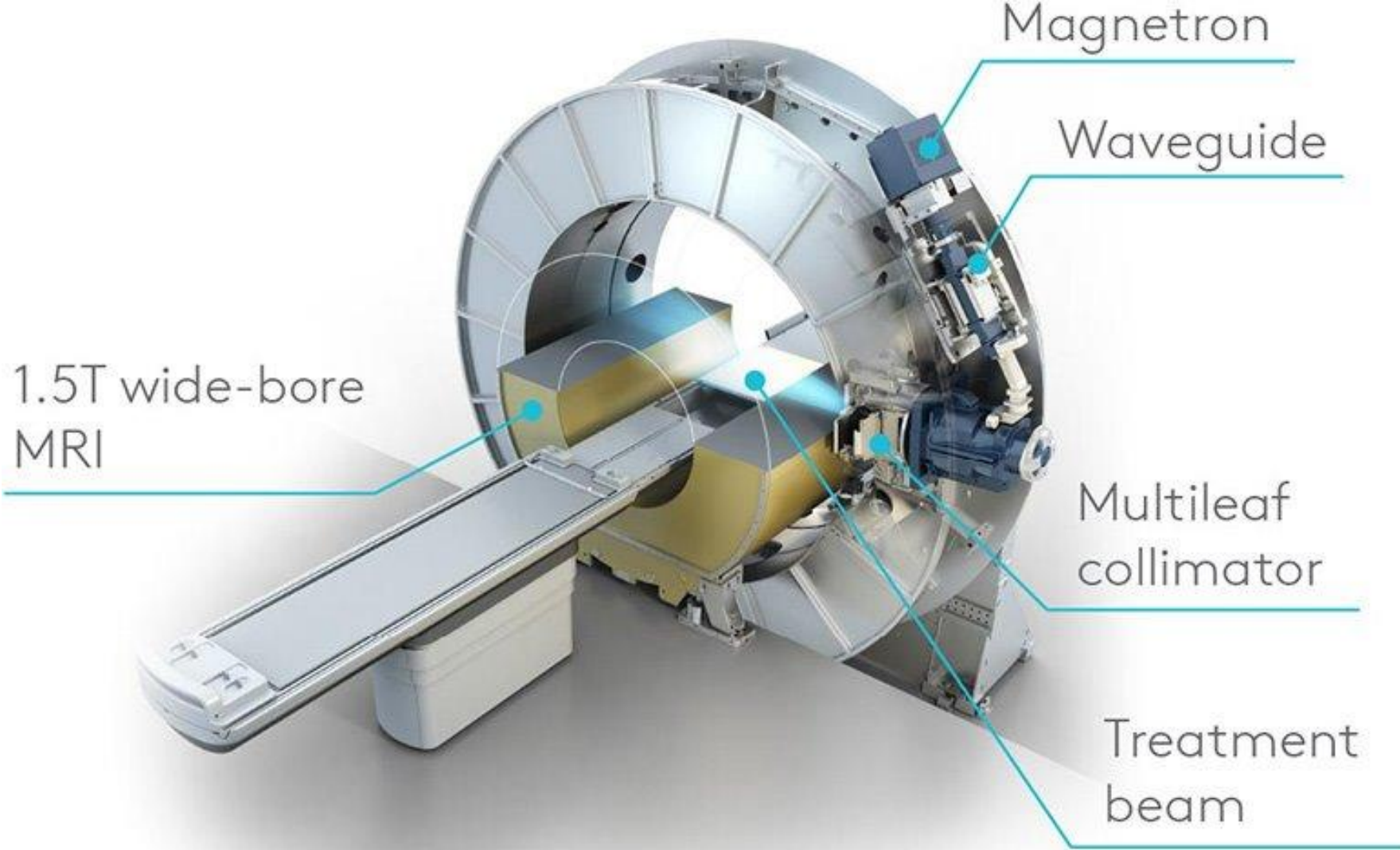
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Agenda

1. Clinical background
2. TrackRAD Grand Challenge overview
3. Out-of-the-box models for object tracking + segmentation
4. Our approach
5. Discussion

MRI-guided linear accelerator (MRI-linac)

- Combine real-time MR imaging with a linear accelerator for radiation treatment
 - Uses 0.35T or 1.5T field (less than diagnostic MR but good enough for soft tissue contrast)
 - Linac = high energy X ray beams
- Adjust X ray beam based on the MRI
 - Precise dose delivery based on individualized tissue geometry
 - Real-time tumor tracking to account for motion during breathing
- Not currently used at Mayo but we are considering getting one



MR-LINAC TIMELINE (LAGENDIJK, RADONC 2022)

1999

MRI guided radiotherapy: a
MRI based linear accelerator
Legendijk JTW and Bakker CJG
2000 Proc. ESTRO Istanbul
19th Annual Meeting
Rad Onc vol 56, suppl. 1, pp
S1-S255

invention

2004



design

2005



Initial exp.

2009



1st prototype

2012



2nd prototype

2014

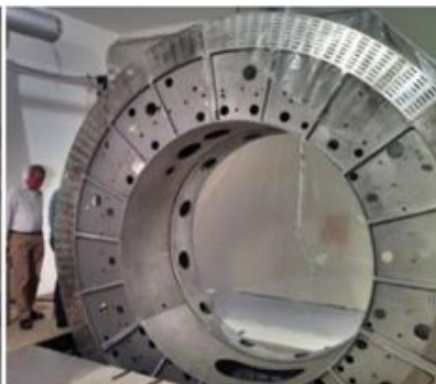


3rd prototype

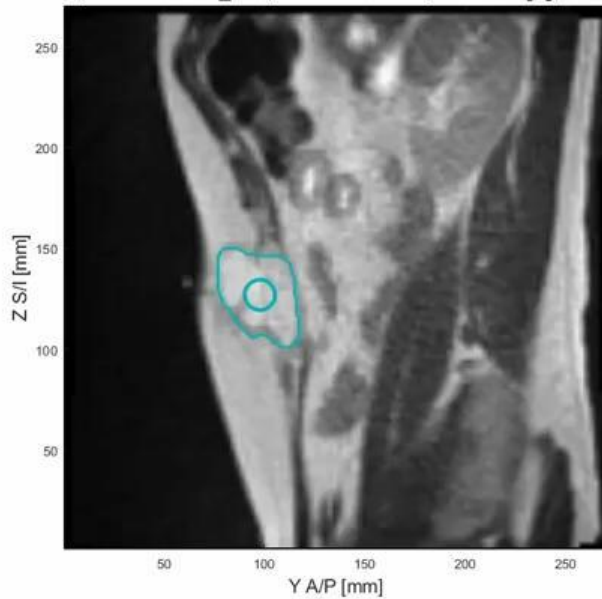
2018



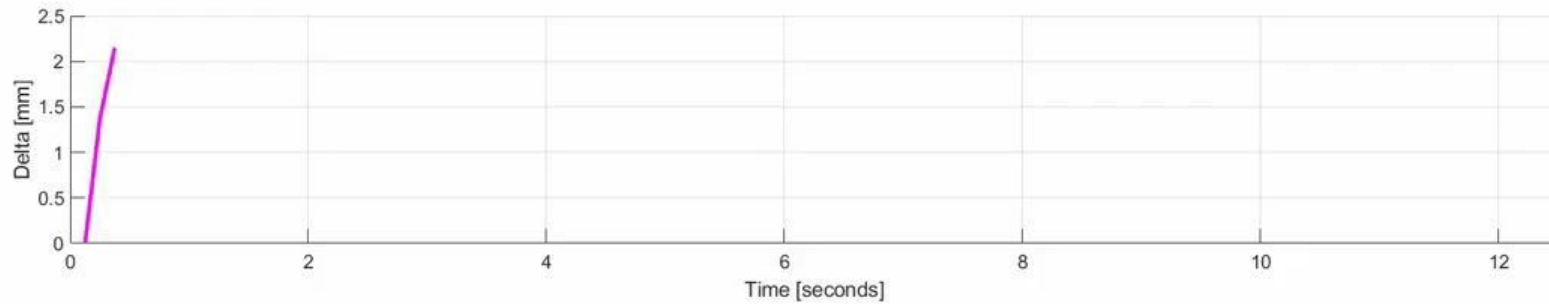
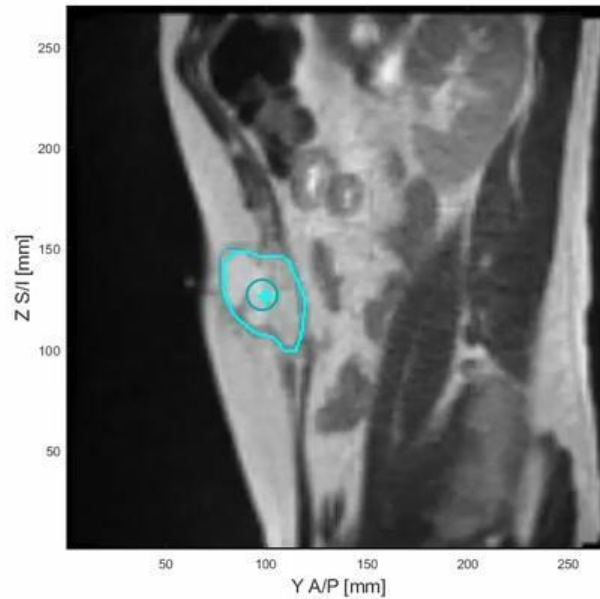
Clinical



Frame 0, Patient ID: A_001, Site: abdomen, B = 0.35 [T], Fs = 8 [Hz]

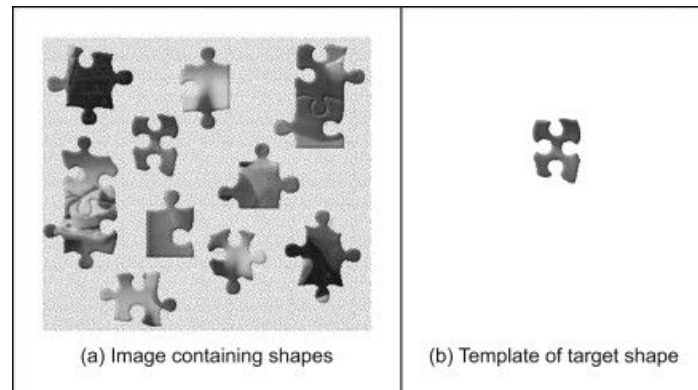
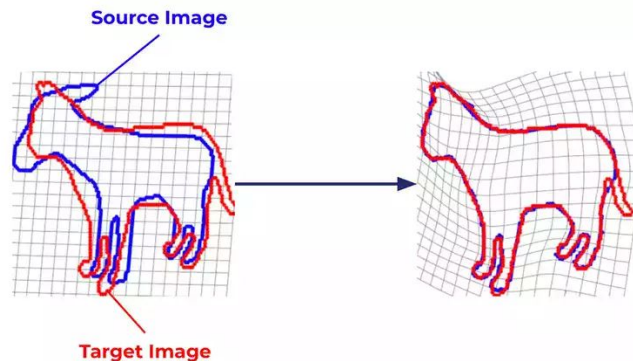


Frame: 3, Delta = 2.2 [mm], $(V, V_0) = (14.9, 15.4)$ [cm²]



Tumor tracking status quo

- Current clinical solutions:
 - Deformable image registration – vector field maps points from frame to frame
 - Template matching – slide a reference image across the target, calculate similarity scores, optimize
- Both struggle with large, non-rigid motion
- Best practice = turn off the beam when “large motion” is detected (gating)
 - Reduces duty cycle, increases treatment time
- MRI adds complexity
 - No standard field strengths, imaging protocols
 - Variance in image quality and artifacts
 - Pixel intensities are relative (vs. CT where 0 intensity = water)



TrackRAD2025 Grand Challenge

- Goal = improve real-time tumor tracking algorithm for MRI-linac
- Input
 - Sequence of 2D MRI frames, 100-200 frames (~10-30s at 8 fps)
 - First frame labeled with segmentation mask
 - Metadata (field strength, frame rate, anatomical region)
- Output = labels for all subsequent frames
- Metrics = image similarity (Dice, etc.), **runtime (<1s/frame on A10 GPU)**
- Dataset
 - From 6 institutions (3 Dutch, 1 German, 1 Australian, 1 Chinese), both 0.35T and 1.5T MRI
 - Train
 - Labeled = 50 patient cases ~ 5000 frames
 - Unlabeled = 477 patient cases ~ **2_800_000 frames**
 - Test = 50 hold-out cases

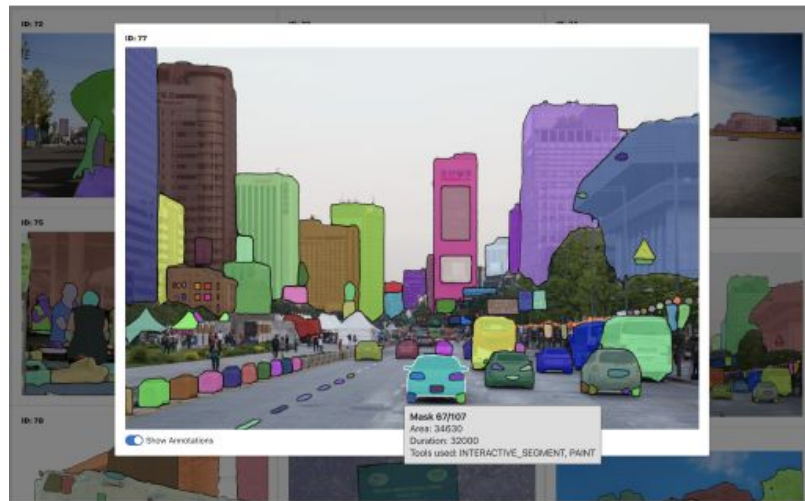
Metrics

<https://trackrad2025.grand-challenge.org/metrics/>

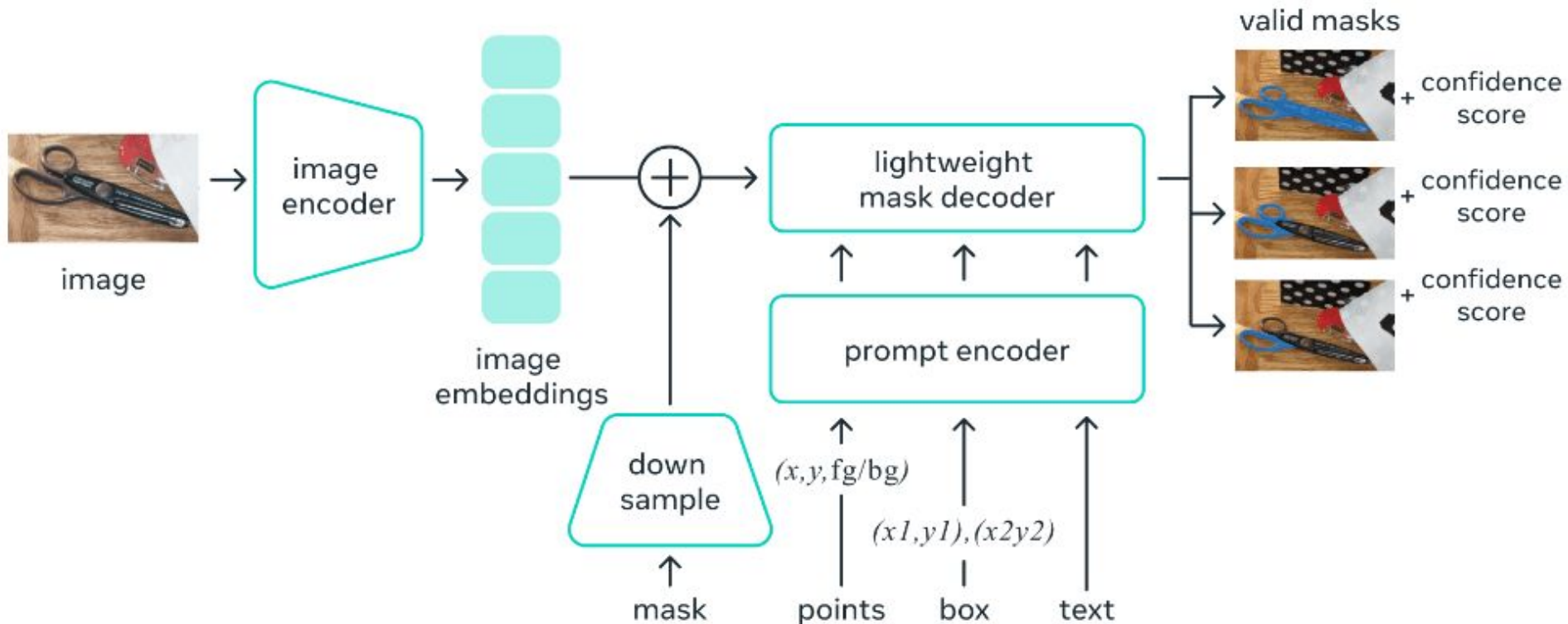
- Dice coefficient (0 to 1)
- P95 Hausdorff distance (pixels)
 - Max distance between a point on the one image to any point on another, over all points
- Mean average surface distance (pixels)
 - Shortest distance between points, over all points
- Euclidian center distance (pixels)
- Dose (0 to 1)
 - Estimates on-target radiation dose accounting for X-ray scatter, tissue-specific fall-off
- Runtime (s)

Segment Anything Model (SAM) (2023)

- Foundation vision model by Meta
- Transformer architecture
- Trained on 11M images + 1.1B segmentation masks
- “Promptable” = use positive/negative points, bounding box
- Good “zero-shot” performance = applied to a task without fine-tuning
- Can be fine-tuned (trained with domain-specific data)
- <https://segment-anything.com/demo>



SAM architecture



SAM2 (2024)

<https://github.com/facebookresearch/sam2>

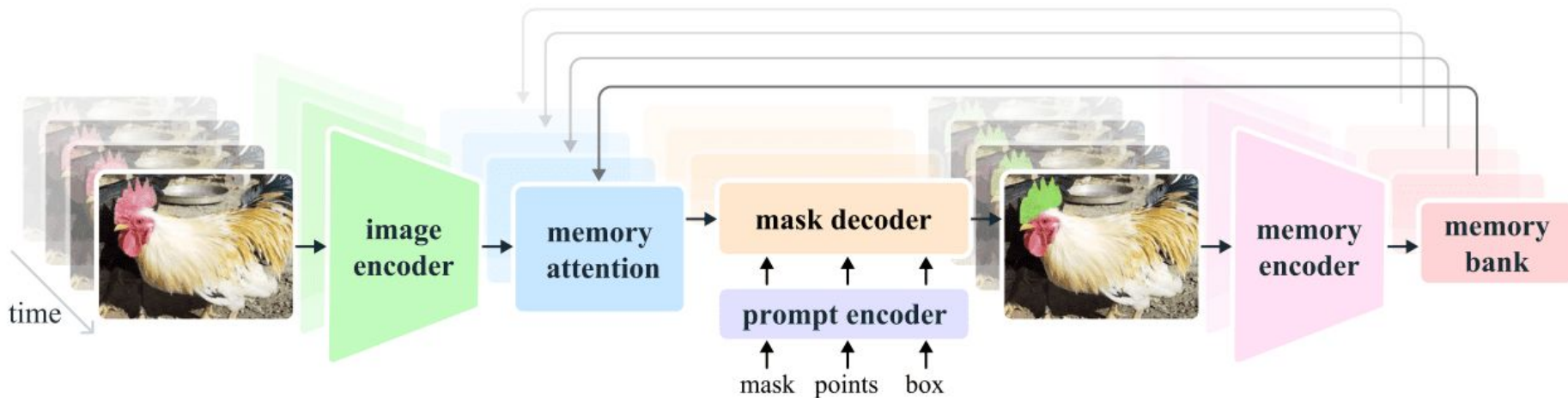
<https://ai.meta.com/sam2/>

- Adds video object segmentation (VOS) functionality
 - **Given one frame + label, infer labels in all subsequent frames**
 - Robust to deformation, object occlusion, entering and leaving the field of view
- Trained on video
 - Training data includes the previous + following 8 frames



SAM2 architecture

- Memory modules allow using masks from previous frames as “prompts”
 - Generalizes the model from image → video object segmentation



SAM2 evaluation

Table 1. Zero-shot predictive accuracy of SAM2.1 checkpoints.

SAM2.1 Check- point	Dice	P95 Hausdorff distance (pixels)	Mean average surface dis- tance (pixels)	Center distance (pixels)	Dose	Runtime per frame (s)
Tiny	0.9072	3.543	1.302	1.593	0.9444	0.0190
Small	0.9158	3.324	1.198	1.390	0.9577	0.0199
Base plus	0.9150	3.510	1.310	1.500	0.9587	0.0240
Large	0.9163	3.266	1.197	1.356	0.9621	0.0356

nnU-Net

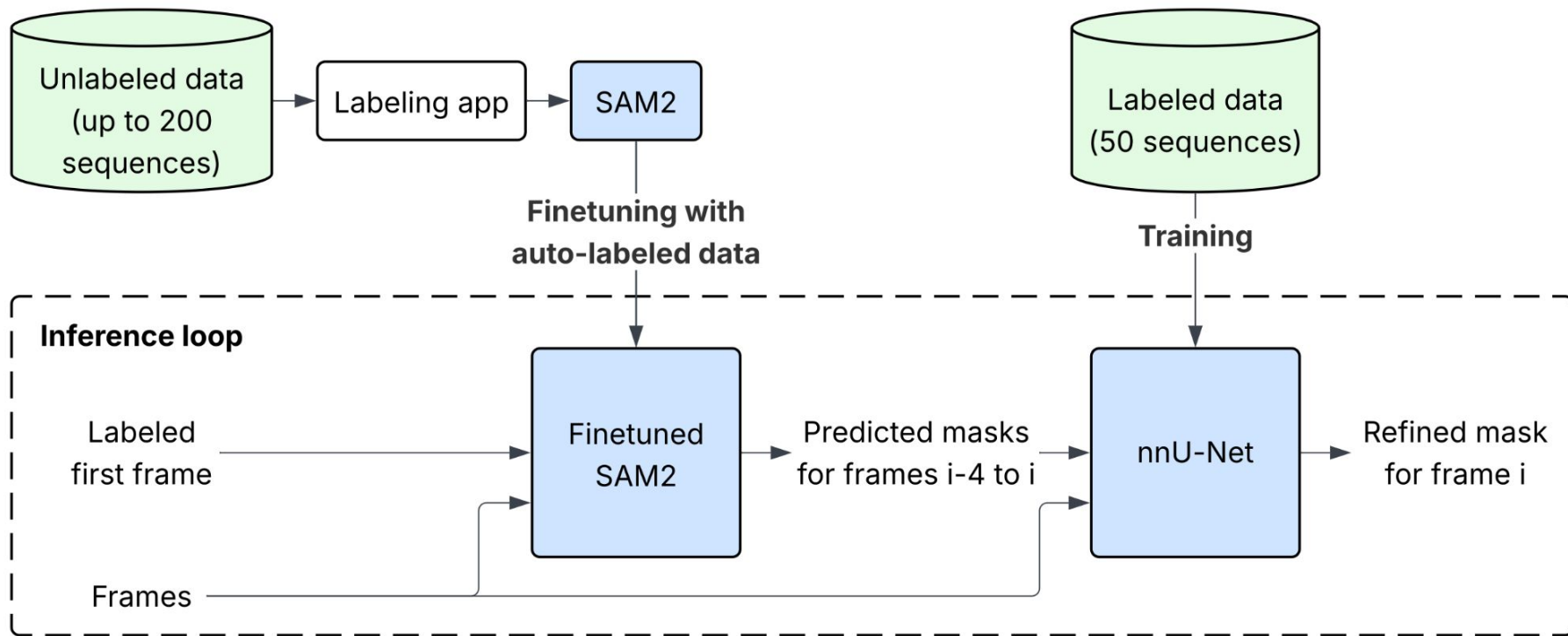
<https://github.com/MIC-DKFZ/nnUNet>

- Auto-configuring U-Net framework (zero-shot auto ML)
- Determines entire pipeline from dataset characteristics
 - Preprocessing, data augmentation, U-Net architecture, etc.
 - Image dimensions, size, modality
- Close to SOTA performance on many segmentation challenges

Modeling plan

1. Use SAM2 for tumor tracking
 - a. Frame sequence + first frame label → Infer subsequent frames
2. Use SAM2 to expedite annotation of unlabeled frame sequences
 - a. Assist in annotating first frames w/bounding box → generate mask
 - b. Infer masks for subsequent frames
3. Fine-tune SAM2 using the labeled data
4. Refine predictions using a lightweight model (U-net)
 - a. Use a sliding window for smoothing + incorporating motion information

Model architecture



Labeling app

<https://huggingface.co/spaces/mayo-radonc/bouncing-target>

Design goals:

- Develop and deploy as quickly as possible
 - Gradio + HuggingFace Spaces + Datasets (app + storage)
- Ease-of-use, rapid labeling
 - Randomly select an image on page load
 - Try to keep everything on one screen without scrolling
 - Reload the page automatically when submitting a label
- Free/cheap as possible
 - Mostly doable!
 - HuggingFace Spaces + Datasets have generous free tiers, ZeroGPU costs \$9/month (but doesn't support private org spaces)

Results – training + validation

- SAM2 fine-tuning
 - 40, 120, and 200 sequence checkpoints
 - Eval on labeled sequences, select best checkpoint
 - Loss curves did not converge during training
- nnU-Net trained on 50 labeled sequences
 - 5-fold cross-validation, average of all folds shown in table

Table 4. Predictive accuracy and runtime after finetuning and postprocessing

Model	Dice	P95 Hausdorff Distance (pixels)	Mean average surface distance (pixels)	Center distance (pixels)	Dose	Runtime per frame (s)
Small zero-shot	0.9158	3.324	1.198	1.390	0.9577	0.0199
Small-120 finetune	0.9195	3.187	1.149	1.306	0.9675	0.0199
Small-120 finetune + nnU-Net	0.9197	3.066	1.116	1.258	0.9706	0.164

Table 5. Percent change in predictive accuracy and runtime relative to zero-shot SAM2

Model	Dice (%)	P95 Hausdorff distance (%)	Mean average surface distance (%)	Center distance (%)	Dose (%)	Runtime per frame (%)
Small-120 finetune	0.404	-4.12	-4.09	-6.04	1.02	0
Small-120 finetune + nnU-Net	0.426	-7.76	-6.84	-9.50	1.35	724

Results – hold-out test set

<https://trackrad2025.grand-challenge.org/evaluation/final-testing/leaderboard/>

- 50 MR sequences, including 2 institutions not included in training data
- Two submissions

Our model (8th place)

"time_per_frame": 0.0453,
"center_distance": 2.50,
"relative_d98_dose": 0.930,
"hausdorff_distance_95": 6.31,
"surface_distance_average": 2.75,
"dice_similarity_coefficient": **0.862**

1st place model

"time_per_frame": 0.0429
"center_distance": 1.47,
"relative_d98_dose": 0.936,
"hausdorff_distance_95": 4.217,
"surface_distance_average": 1.66,
"dice_similarity_coefficient": **0.891**

Limitation – data quality

- Image annotators were not trained
- Vast majority of training data was generated by the model
- No validation was done, manual or automated (e.g., does the object disappear, large change in volume, etc.)

Table 2. Predictive accuracy of SAM2.1 after finetuning. Checkpoints are named for the number of MR sequences used for training.

SAM2.1 Check- point	Total frames	Dice	P95 Hausdorff Distance (pixels)	Mean aver- age surface dis- tance (pixels)	Center distance (pixels)	Dose
Small	0	0.9158	3.324	1.198	1.390	0.9577
Small-40	9960	0.9192	3.212	1.157	1.325	0.9624
Small-120	35,531	0.9195	3.187	1.149	1.306	0.9675
Small-200	92,441	0.9187	3.231	1.161	1.331	0.9613

Unsuccessful methods

- Starting with MedSAM2 (<https://github.com/bowang-lab/MedSAM2>)
- Fine-tuning SAM2 with another MRI dataset (Duke Liver) – out-of-domain
- Training specialized models based on metadata (anatomical region, field strength, FPS) and routing at inference time
- MRI-specific data augmentation for SAM2 and nnU-Net training (motion artifacts, intensity variation, etc.)
- nnU-Net using residual encoder U-Net – overfit
- Longer sliding windows (10, 15 frames)

Personal reflections

- For the inexperienced, hard to decide on next step to improve performance
 - Data quality?
 - Data augmentation?
 - More data?
 - Hyperparameter tuning?
 - Model architecture?
- Should have tried harder to find trained image annotators (apparently a radiology trainee was available)

Future work

- Lightweight browser-based image annotation/visualization tools
- Active learning via automated data validation + flagging images for relabeling
- SAM3 (late 2025 or 2026?)
- See what the winning teams did (paper will come out in a few months)

References

<https://github.com/mzhu22/trackrad2025>